

## **Sustainable Energy in Constructing Buildings**

Globalization has led to the increased urbanization in many countries across the world, which means that there is a lot of building taking place. In the current world, 42 percent of the electricity use is attributable to the building sector (UNIDO, 2017). This percentage is high in comparison to any other industry as verified by the statistics given by the International Energy Agency (IEA). Increasing urbanization, especially in the developing countries such as Africa, consequently leads to the increase in the number of buildings and their sizes. For this reason, the demand for electricity among other types of energy needed for construction also increases significantly. This paper compares and contrasts two articles to get an insight on sustainable energy in constructing buildings.

The article written by the United Nations Industrial Development Organization (UNIDO) focuses mainly on sustainable energy in the construction of buildings in developing countries. The report emphasizes the point that these developing countries are the ones that consume a lot of energy in their development due to the need for urbanization. For instance, the rate of urbanization in Africa is the highest in the world accounting for 3.5 percent annually (UNIDO, 2017). For this reason, they consume a lot of energy as compared to the other developed countries like the United States of America among other nations.

World Business Council for Sustainable Development (WBCSD) writes the article, which focuses on energy efficiency worldwide. Both UNIDO and WBCSD (2017) identify the benefits that are likely to emerge from the energy efficiency in the building sector which include lower maintenance requirements, enhanced property value, and improved comfort. The barriers that these two articles identify regarding sustainable energy include both the organizational, behavioral, and financial obstacles. In an attempt to eliminate these challenges, the building sector should be able to adopt holistic and integrated methods among its stakeholders. This approach encourages interdependence and makes sure that there is accountability and the stakeholders share their responsibilities towards a more sustainable energy consumption.

The building sector should also work towards making the stakeholders involved in the development of the building value it more and transform their behavior by motivating them and educating them on the importance of energy efficiency in the construction of buildings. UNIDO in the article identifies three main approaches that can aid in the development of sustainable energy in the building sector. These methods include cutting the energy demand for the building by utilizing more energy efficient equipment. Producing the energy locally using renewable resources can provide surplus energy and use it elsewhere.

On the contrary, the article done by the WBCSD identifies seven approaches to promoting sustainable energy in the construction sector. These strategies include reducing the demand for heating, reducing the cooling demand, reducing the energy needed for ventilation, and reducing the use of energy for lighting. It entails reducing the energy utilized for heating water, reducing the electricity use for the office appliance, and good housekeeping by solving behavioral problems or finding people solutions (WBCSD, 2017).

In conclusion, both articles emphasize on the fact that the reduction in the energy consumed by the appliances and equipment used for the construction of the buildings provide new opportunities for a sustainable energy use. The enhancement in building designs as outlined in the UNIDO article also encourages energy savings. The building sector should, therefore, embrace these changes rather than resist it to allow a more sustainable energy in the construction of buildings across the world.

## References

- UNIDO. (2017). *Sustainable energy in building construction*. United Nations Industrial Development Organization.
- WBCSD. (2017). *Energy efficiency in buildings: Strategies for sustainable development*. World Business Council for Sustainable Development.